



DOE CSGF Application - Traditional Track

Name: Olivia Asher

Applicant Information

First name	Middle name	Last name
Olivia	Allison	Asher
Previous names (ex: maiden name)		
Email		

Citizenship

Country of Birth:	United States
U.S. Citizen:	Yes
If no,	
• Country of Citizenship:	NY
• Lawful Permanent Resident:	----
If yes,	
• LPR Number:	
• Port of Entry:	New York

Name: Olivia Asher

Current Mailing Address

Street address 1

Street address 2

City

State

Zip code

Home hone

Cell phone

Work phone

Address effective through (m/d/y):

After this date, all correspondence will be sent to the permanent address listed below unless otherwise requested.

Notify the Krell Institute if your address changes after the application has been submitted.

Permanent Address

Zi code

References

List at least three persons familiar with your academic preparation and your technical abilities.
Please have these individuals mail the reference forms directly to Krell Institute.

	<i>Title</i>	<i>First name</i>	<i>Last name</i>	<i>Institution</i>	<i>Email</i>	<i>Status</i>
1.				Department of Genetics, University of Georgia		Submitted
2.				Institute of Systematic Botany, NYBG		Submitted
3.				Department of Biology, Lehman College, CUNY		Submitted
4.				Royal Botanic Gardens Victoria		Submitted

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Academic Status

Current Academic Status: **Undergraduate Student**

Have you completed any academic credit towards your computational science/engineering doctoral degree? **No**

If yes, how many terms have you completed? (exclude summer) ----

Official transcripts from every listed institution are a required component of the application including your Fall 2020 transcript, if applicable.

Doctoral Institution (Institution where you plan on completing your computational science and engineering doctorate or first choice doctoral university):

<i>Institution</i>	<i>Start Date</i>	<i>Expected End Date</i>	<i>Department</i>	<i>Academic Discipline</i>	<i>GPA</i>
University of Georgia	08/2021	05/2025	Institute of Bioinformatics	Bioinformatics	

Department Chair at Doctoral Institution:

<i>First Name</i>	<i>Last Name</i>	

Other Doctoral Institution Choices (Answer only if not currently at doctoral institution)

			<i>Department Chair Information</i>	
<i>Institution</i>	<i>Department</i>	<i>Academic Discipline</i>	<i>Name</i>	<i>Email</i>
Purdue University	Interdisciplinary Life Sciences	Computational and Systems Biology		
Cornell University	Computational Biology	Computational Biology		

Higher Educational History (All university/colleges attended and degrees obtained with the exception of the doctoral degree listed above):

<i>Institution</i>	<i>Start Date</i>	<i>End Date Expected or Actual</i>	<i>Department</i>	<i>Academic Discipline</i>	<i>Degree</i>	<i>GPA</i>
Macaulay Honors College at Lehman College, CUNY	08/2017	05/2021	Biology	Bioenvironmental Science	Bachelors	4.0
					None	
					None	
					None	
					None	

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Graduate Advisor

The graduate advisor is the person from the preferred institution **who views and approves the Program of Study.**

Last Name

te r., s., Professor, ...)

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Cit

Zip Code

Telephone

Fax

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Program of Study

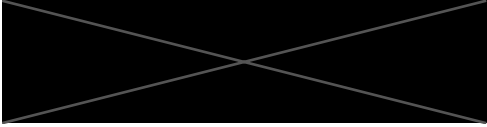
Listed are the courses in science and engineering, applied mathematics, and computer science that you agreed to take on your proposed Program of Study.

University: University of Georgia

Course number	Course Title	Credit hours	Term and Year	Grade	Academic Level
Science/Engineering					
GENE 8960	Genetics of Yeast and Filamentous Fungi	3S	Spring 2023		G
PATH 6200	Mycology	3S	Fall 2022		G
Mathematics and Statistics					
MATH 6700	Qualitative Ordinary Differential Equations	3S	Spring 2023		G
MATH 6780	Mathematical Biology	3S	Fall 2023		G
Computer Science					
CSCI 6130	CUDA programming on CPUs	3S	Spring 2022		G
CSCI 6140	Numerical Methods	3S	Fall 2022		G

I have read this program of study and affirm that, in my opinion, it satisfies the fellowship program requirements. This POS has been approved by my advisor, , and I understand that, if offered a fellowship, my advisor and I are required to sign this page and send it to the Krell Institute.

Student's signature _____ Date _____

Graduate Advisor 
Graduate Advisor _____
Graduate Advisor _____ Date _____

Krell Institute (Office use only) _____

Krell Institute, Attn: DOE CSGF Coordinator
1609 Golden Aspen Drive, Suite 101, Ames, IA 50010
Phone: 515-956-3696, Fax: 515-956-3699, csgf@krellinst.org

Name: Olivia Asher

Course Description

GENE 8960: Genetics of Yeast and Filamentous Fungi

Classical genetics and molecular biology of budding yeast, fission yeast, and model filamentous fungi. Includes life cycle, cell cycle, cytoskeleton, mating types, morphogenesis, pathogenesis, mutant screens, and cloning strategies.

PATH 6200: Mycology

The biology, ecology, genetics, morphology, and taxonomy of fungi and groups of organisms traditionally classified with the fungi.

MATH 6700: Qualitative Ordinary Differential Equations

Transform methods, linear and nonlinear systems of ordinary differential equations, stability, and chaos.

MATH 6780: Mathematical Biology

Mathematical models in the biological sciences, systems, phase-plane analysis, diffusion, convective transport, bifurcation analysis. Possible applications will include population models, infectious disease and epidemic models, acquired immunity and drug distribution, tumor growth, and analysis of arterial flow dynamics.

CSCI 6130: CUDA programming on CPUs

GPUs are high-performance many-core processors. Students will learn massively parallel programming using CUDA C and develop parallel algorithms to solve real problems. Topics include parallel computing, GPUs architecture, threads, performance issues, and floating point representation.

CSCI 6140: Numerical Methods

Numerical methods and computing. Topics include: computer arithmetic; numerical solutions of nonlinear equations; polynomial interpolation; numerical differentiation and integration; numerical solutions of systems of linear equations, initial and boundary value problems, systems of ordinary differential equations, spline functions, and the method of least squares.

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Other Planned Courses

Listed are the other courses you plan to take that you believe are particularly pertinent to your proposed or current research in the areas of Mathematics, Science and Engineering, and Computer Science.

No courses have been added.

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Course Description

No courses have been added.

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Completed Courses

Please list up to six courses you have completed that are particularly pertinent to your proposed or current research in the areas of Mathematics, Science and Engineering, and Computer Science. Please do not list entry level science/engineering or mathematics courses like Calculus I.

Course number	Course Title	Credit hours	Term and Year	Grade	Academic Level
BIO 240	Biostatistics	3S	Spring 2019	A	U
BIO 330	Plant Physiology	4S	Spring 2020	A	U
BIO 331	Experimental Microbiology	4S	Fall 2020	A	U

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Course Description

BIO 240: Biostatistics

Course was taken at Lehman College. Data analysis and construction of tables, figures and paragraphs suitable for publication in biological journals. Experimental design informed by power analysis of preliminary data. Introduction to randomization, bootstrap and Monte Carlo in biological research; and coverage of traditional topics in biostatistics. Computer programming of biological algorithms in JSL (SAS) and R.

BIO 330: Plant Physiology

Course was taken at Lehman College. Consideration of the major physiological processes of plants, with special emphasis on water relations, inorganic nutrition, photosynthesis, metabolism, and hormonal relationships. Laboratory studies consist of physiological experiments with living plants.

BIO 331: Experimental Microbiology

Course was taken at Lehman College. Consideration of major groups of microorganisms with emphasis on metabolic activities, chemical composition, ultrastructure, genetics, ecology, and phylogenetic relationships. Basic and advanced microbiological techniques.

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Research Statements

This information is vital to the overall evaluation of your application.

Field of Interest and the Role of Computational Science

- a. In terms a general audience would understand, describe an important, outstanding scientific or engineering challenge in your field of interest where computational science can play an important role. (1/3)*
- b. Describe the particular science or engineering problem that you would like to pursue in your research. What would be the impact on the field and/or on science, engineering and/or society in general if this challenge could be successfully addressed? (2/3)*

In the dark underbelly of every ecosystem networks of tiny fungal filaments called “hyphae” snake through the soil. Some of the hyphae penetrate the roots of plants growing near them and form circular colonies inside of plant tissue called “arbuscules.”

Plants and the fungi which form these arbuscules (arbuscular mycorrhizal fungi or AMF) form a symbiotic relationship where the plant provides the fungus with carbon and the fungus provides the plant with nutrients like phosphorous and nitrogen. This relationship evolved over 400 million years ago, and around 90% of plants have the ability to form symbioses with AMF.

One of the main goals of agriculture is to maximize crop yields while maintaining the viability of the land. AMF help provide plants with the nutrients they need to grow, and if the power of this symbiosis could be harnessed to maximize plant growth and health, farmers could increase crop yields while decreasing fertilizer use. The potentials of AMF are currently underexploited due to lack of knowledge on these organisms and their relationship with plants.

Previously conducted research on AMF-plant symbiosis has focused on the interaction between one plant type and one AMF species. However, in the real world, hundreds of species of AMF fungi may associate with one plant. Understanding the complexities of these fungus-fungus and plant-fungus interactions would allow for development of agricultural tools to enhance beneficial symbioses and increase crop yields. My research interest is to use computational modeling tools to both understand the dynamics of this complex system and to make predictions about how to best manipulate conditions to maximize the effectiveness of plant-AMF collaboration.

Unraveling the interactions of AMF-plant symbiosis will allow for benefits to agriculture and, in addition, the techniques used in this process could be used to unravel other biologically interesting systems. Crop plants are not the only plants which form fungal symbioses, and understanding how AMF affect crops could help us understand how these symbioses impact natural ecosystems as well.

Citations (Optional)

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Use of Computational Science in your research

- a. *What is the most complex calculation you have run on a high-performance machine as part of your research experience? Or if you haven't run a high-performance computing system, tell us about the most complex computational problem you have tackled. (1/2)*
- b. *Imagine if you were given access to resources 100 times more powerful than what you have access to. What would that enable you to do, and what do you perceive the mathematical and computer science challenges to be? (1/2)*

While I have not previously engaged in high performance computing, I am surrounded by researchers at the University of Georgia (UGA) who are doing so. My current research project, which I started through a Research Experience Undergraduate internship at UGA this summer, is to develop a model of AMF/plant symbiosis which will eventually be expanded to use advanced computing resources. The system of equations for the preliminary model are based on the Lotka-Volterra model of species competition and were expanded to include both fungus-plant collaboration and fungus-fungus competition. The next step is to fit the model by ensemble methods to real molecular data on a high-performance computing cluster using hybrid CPU/GPU nodes. The group that I work with were the first developers of these ensemble methods in systems biology derived from statistical physics [1]. The modeling project is part of a research initiative funded by the Department of Energy's Systems Biology to Advance Sustainable Bioenergy Crop Development (DE-FOA-0002214) under the Award DE-SC0021386.

The main challenge of modeling, especially advanced computational modeling, is ensuring that the model reflects the real world. I will collaborate with experimental biologists and compare my model to their results and to verify its accuracy.

I have also conducted phylogenetic research on fungi at the New York Botanical Garden and the Royal Botanic Garden, Victoria, Australia. The purpose of this research was to unravel the evolutionary history of understudied genera and identify new species. I used maximum likelihood and Bayesian methods to analyze genetic data and to create phylogenies. Running phylogenetic analyses on large datasets takes hours. If I could use high powered computing methods for this research, I would analyze larger datasets and full genome sequences rather than the short reads and small datasets I currently use. Larger datasets enable broader understanding of study groups, but it would be more challenging to find sequencing errors, especially since errors in understudied groups may be unique to the group. To mitigate this challenge, I would test many error correction algorithms on small datasets before applying them to the large dataset

Citations (Optional)

1 Battogtokh, D., Asch, D. K., Case, M. E., Arnold, J., & Schuttler, H. B. (2002). An ensemble method for identifying regulatory circuits with special reference to the qa gene cluster of *Neurospora crassa*. *Proceedings of the National Academy of Sciences*, 99(26), 16904-16909.
doi:10.1073/pnas.262658899

Program of Study

Describe how the courses listed in your planned program of study would help prepare you to address the challenges you have described in questions 1 and 2. Discuss your rationale for choosing these courses.

Computational biologists are liaisons between the world of pure computation and the organic and complicated world of observation-based and experimental biology. To be effective researchers, computational biologists must understand both of these worlds link them together via their research. I have designed my plan of study with the goal of being equally grounded in biology and computation in mind.

The two science courses I listed, Genetics of Yeast and Filamentous Fungi and Mycology will enable me to expand upon my knowledge of the focal organisms of my thesis (AMF) and facilitate communication with experimental collaborators. Also, learning the biology and genetics of fungi will help me think of creative ideas about how to model and explain their biology in a computational manner. To ensure I am well-prepared to make connections between the biological world and the world of mathematics, I will take Mathematical Biology.

My modeling work heavily relies on ordinary differential equations (ODE). All of my current knowledge on ODEs comes from research and introductory calculus. To develop into a fully-fledged computational biologist working in modeling, I must ensure I have a deep understanding of my methods. I have chosen two courses which specifically focus on ODEs, Qualitative Ordinary Differential Equations, and Numerical Methods. The first course is a mathematics class and will ensure I am well versed in the qualitative features of my model. The second course, is a computer science course which will allow me to learn improved methods of coding ODEs and their ODE solvers, knowledge which will be invaluable to my research goals.

Finally, I plan to take CUDA programming on CPUs. My current model consists of only one system of ODEs, but as I move forward on this I plan to engage in model averaging over 40,000 models using ensemble methods. Taking CUDA programming on CPUs will allow me to understand how to run these computations effectively and develop a useful model explaining the complex interactions between hundreds of species of arbuscular mycorrhizal fungi and their plant host.

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Programming Languages and Models

List (four at most) the programming languages and programming models with which you have experience. Provide a sentence that describes how you use them.

1. Programming Language/Model: **R**

Ancestral state reconstruction

2. Programming Language/Model: **MATLAB**

Modeling using ODEs

3. Programming Language/Model: **RAxML, MrBayes**

Phylogenetic analysis

4. Programming Language/Model:

What are the programming languages that you intend to use in your research?

MATLAB, CUDA C (C++), Open CL

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List of Publications

Papers

Asher, O.A. & Lendemer, J.C. (2021). Molecular data provide a new perspective on the evolution of reproductive modes and biogeography in the lichen genus *Platismatia* (Parmeliaceae, lichenized ascomycetes). [in prep]

Asher, O.A. & Davoodian, N. (2020). The Boletales fungi of New York City, Rhodora. [in press]

Davoodian, N., Hosaka, K., Raspe O., Asher, O.A., Franck, A.R., De Kesel, A., Delaney, T.P., Ammirati, J.F., Nagasawa E., B. Buyck & Halling, R.E. (2020). Diversity of *Gyroporus*

(Gyroporaceae, Boletales): rpb2 phylogeny and three new species. *Phylotaxa* 434(3), 208-218.

Duncan, N., Asher, O.A., Weckel, M., Nagy, C., C. Henger & Yau, F. (2020). Baseline diet of an urban carnivore on an expanding range front. *Journal of Urban Ecology*, 6(1),

<https://doi.org/10.1093/jue/juaa021>.

Asher, O.A. & Lendemer, J.C. (2018). *Lecanora caperatica* (Lecanoraceae, lichenized ascomycetes) a new sorediate species widespread in eastern North America. *The Bryologist*, 121(3), 306--323.

Talks

University of Georgia Research Experience Undergraduate Symposium 2020 (Online) -- Who Will Survive? Ecological Modeling of Nutrient Exchange Between Plants and Arbuscular Mycorrhizal Fungi. <https://youtu.be/DVKpYmVcHTA>

Northeast Natural History Conference 2017 (Cromwell, CONN) -- Spatial Variation in the Diet of New York City Coyotes

Posters

Mycological Society of America Meeting 2019 (Minneapolis, MN) -- Phylogenetic Analyses of the Genus *Platismatia* (Parmeliaceae, Lichenized Ascomycetes) Shed New Light on Species Relationships

Northeast Natural History Conference 2019 (Springfield, MA) -- Phylogenetic Analyses of the Genus *Platismatia* (Parmeliaceae, Lichenized Ascomycetes) Shed New Light on Species Relationships

Student Undergraduate Research Conference 2019 (New York, NY)-- Phylogenetic Analyses of the Genus *Platismatia* (Parmeliaceae, Lichenized Ascomycetes) Shed New Light on Species Relationships

Lehman Student Scholars Conference 2019 (New York, NY) Phylogenetic Analyses of the Genus *Platismatia* (Parmeliaceae, Lichenized Ascomycetes) Shed New Light on Species Relationships

Student Conference on Conservation Science 2016 (New York, NY) Diet of New York City Coyotes

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Laboratory and Research Experience/Other Employment

Begin with current or most recent employment. Please include employer, dates employment started and ended, position, and nature of work.

Laboratory Research:

University of Georgia, 05/2020-present, Research Experience Undergraduate Intern, computational biology research.

New York Botanical Garden, 01/2018-present, Lichenology and Mycology Research Assistant, taxonomy, phylogenetics, and systematics research.

Royal Botanic Garden, Victoria, Australia, Summer 2019, Mycology Research Assistant, taxonomy, phylogenetics, and systematics research.

American Museum of Natural History, Department of Mammalogy, 09/2015-08/2017, Science Research Mentoring Program Intern, researched diet of New York City coyotes.

Employment:

Lehman College, Instructional Support Services, 02/2019-present, Genetics Tutor (College Assistant), assist students individually and in groups in studying genetics.

American Museum of Natural History, Lang Science Program, Summer 2017, Teaching Assistant, classroom management and teaching on diet of New York City coyotes.

Academic Awards and Honors - *Include undergraduate and graduate honors (if applicable).*

Macaulay Honors College, Full Merit Full-Tuition Scholarship (Four Year), 08/2017, 2018, 2019, 2020, 05/2021.

Lehman College, Presidential Scholar (Academic honor), 2017, 2018, 2019.

Macaulay Honors College, Goldsmith Scholar (Academic honor), 2019, 2020, 2021.

Torrey Botanical Society, Undergraduate Student Research Fellowship (\$1,000), 2019.

Macaulay Honors College, Enhanced Opportunities Fund (\$5,000), 2019.

Mycological Society of America, Undergraduate Research Award (\$500), 2019.

Con Edison Inc., Con Edison Endowed Scholarship (\$5,000), 2019.

National Institute of Health, Research Initiative in Scientific Enhancement (RISE) (\$3,040) 2019, 2020, 2021.

Extracurricular Activities - *Include technical societies and service organizations.*

Founding president of the Lehman College Flora, Fauna, Fungi Trails Club 2019-2020.

Member of the Lehman College Division III swim team, 2019-2020.

Volunteer with New York Cares 2018-2020.

Founder and author of thesciencenotebookblog.wordpress.com 2015-present.

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Additional Comments



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DOE CSGF and Other Fellowships

The information that you provide will allow us to target our advertising more effectively. This information is confidential and is not used in review of the fellowship application.

1. Have you applied to other fellowship programs?

- ☐ DOE NNSA SSGF
- ☐ NSF
- ☐ DOD
- ☐ University-sponsored Names of fellowships:
- ☐ Other Names of fellowships:

2. How did you find out about the program?

- ☐ DOE CSGF poster
- ☐ DEIXIS, DOE CSGF annual publication
- ☐ Attended DOE CSGF talk
- ☐ Advertisement Name the source:
- Word of mouth from
 - ☒ faculty
 - ☐ student
 - ☐ administrator
- ☐ Laboratory staff
- ☐ Institutional announcement
- ☐ Conference or meeting Name:
- ☐ World Wide Web List URL:
- ☐ Other Explain:

A llicant Demo ra hics

